Final Project - Deployment Proposal for PromisedLand.com at Independence

7.2 Prove - IT235.001 - Cloud Server Administration

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Overview

This proposal outlines the strategic deployment plan for PromisedLand.com's new data center in Independence, Missouri. This facility will serve as a secondary data center, supporting our hybrid call centers across the US and Canada. The proposal details key elements of the deployment, including infrastructure, software, security, and management, ensuring a robust, scalable, and secure environment. A demonstration video showcasing the implementation of critical services is embedded within this document.

Part A

1. Managing Roles and Features

To support network services, authentication, and data security, the following Windows Server 2022 roles and features will be deployed:

- Active Directory Domain Services (AD DS): Centralized authentication and identity management.
- Domain Name System (DNS): Name resolution services, including Primary and Secondary.
- Dynamic Host Configuration Protocol (DHCP): Automated IP address allocation and lease management.
- File and Storage Services: Secure and redundant file sharing using Distributed File System (DFS).
- Internet Information Services (IIS): Hosting internal applications, portals, and web-based services.
- Hyper-V: Virtualization of critical workloads to improve efficiency and disaster recovery.
- Network Policy Server (NPS): RADIUS authentication for secure remote access.
- Group Policy Management: Enforcing security policies, access controls, and compliance measures.

Why: This role-based deployment model ensures scalability, security, and streamlined management, reducing administrative overhead while optimizing performance. By assigning specific roles and features, we can fine-tune our server performance, reduce system overloads, and create a symphony of reliability.

2. Cloud Concepts + Virtualization

To maximize resource utilization and scalability, we will leverage virtualization technologies. This will allow us to run multiple virtual servers on a smaller number of physical machines, reducing hardware costs and energy consumption.

- Cloud Integration (Azure Services)
  - Azure Active Directory (Azure AD): Extending authentication across cloud and on-premises environments.
  - Azure Backup: Securing mission-critical data with automated, encrypted cloud backups.
  - Hybrid DNS Configuration: Ensuring seamless resolution of cloud-hosted services.
- On-Premises Virtualization (Hyper-V)
  - Server Consolidation: Running multiple virtual machines (VMs) on fewer physical servers.
  - High Availability and Fault Tolerance: Implementing failover clustering.

Why: Cloud and virtualization technologies offer a canvas for flexibility and cost-efficiency, allowing us to allocate resources on-demand and create virtual realms for testing and development.

3. Managing IP Addressing

We will implement a structured IPv4 addressing plan, potentially transitioning to IPv6 in the future. DHCP will be used to dynamically assign IP addresses to devices, simplifying network administration and preventing conflicts. Subnetting will be employed to segment the network for security and performance optimization.

- Subnetting Strategy: VLANs for servers and workstations.
- DHCP Configuration: Reservations and exclusions for critical devices.
- IP Address Management (IPAM): Automated tracking and management of network resources.

Why: A well-managed IP addressing scheme is the key to avoiding conflicts, simplifying network management, and ensuring devices communicate effortlessly.

4. Files and Directories

We will establish a clear and organized file and directory structure to ensure easy access to information and data while maintaining appropriate security.

- Access-Based Enumeration (ABE): Restricting file visibility based on user permissions.

- DFS (Distributed File System): High-availability file sharing across multiple locations.
- NTFS Permissions and Audit Policies: Ensuring secure data access and regulatory compliance.

Why: Organizing files and directories systematically enhances security, facilitates efficient data retrieval, and ensures only authorized individuals access sensitive information.

5. Hardware and Software Installation

The hardware and software selection will be based on performance requirements, scalability needs, and compatibility with existing systems. We will procure servers, network equipment, and other hardware from reputable vendors. Software installation will follow best practices, including proper licensing, configuration, and testing. A detailed inventory of all hardware and software will be maintained for asset management and support.

- Server Specifications: Enterprise-grade CPU, RAM, RAID storage, and redundant power supplies.
- Windows Server 2022 Datacenter Edition: Providing robust security and virtualization capabilities.
- SIEM (Security Information and Event Management): Centralized threat detection and monitoring.

Why: Utilizing the latest hardware and software ensures optimal performance, minimizes compatibility issues, and grants access to the newest features and security enhancements.

6. DNS Implementation + Management

Picture a directory that translates names into locations. DNS (Domain Name System) is our directory, translating domain names into IP addresses. We'll implement and manage DNS for reliable name resolution and efficient network traffic routing.

- Primary and Secondary Zones: Load balancing and fault tolerance.
- DNSSEC Implementation: Protecting against DNS spoofing and cache poisoning attacks.

Why: Proper DNS implementation guarantees easy access to network resources using domain names, enhancing user experience and simplifying network management.

7. Users & Groups + Management

Effective user and group management is crucial for access control and security. We will create user accounts with **strong passwords** and assign them to **appropriate groups** based on their roles and responsibilities. Group Policy will be used to enforce security policies and manage user settings. Regular audits of user accounts and group

memberships will be conducted to ensure compliance and identify any potential security risks.

Why: Effective user and group management fortifies data security, prevents unauthorized access, and streamlines the administration of user privileges.

8. Active Directory + Group Policy Objects

Active Directory will be the central directory service for managing users, computers, and other network resources. **Group Policy Objects** (GPOs) will be used to enforce security policies, configure **user settings**, and deploy software. This centralized management will simplify administration and ensure consistency across the environment. We will create an **organizational unit** (OU) structure that mirrors the company's organizational chart for efficient management.

Why: AD and GPOs provide a centralized command center to manage and enforce policies, ensuring consistent security settings and configurations.

9. Managing Servers in a Hybrid Model

We will utilize System Center for managing on-premises servers and Azure Management tools for cloud resources. This integrated approach provides a single pane of glass for monitoring, managing, and automating server operations across the hybrid cloud.

- ➔ Azure AD Connect: Seamless cloud identity synchronization.
- ➔ Load Balancing: Distributing workloads between on-premises and cloud.
- ➔ Hybrid Backup Strategy: Combining local and cloud-based redundancy.

Why: A hybrid model offers control and security of on-premises servers with the flexibility and scalability of cloud services, creating a coordinate ecosystem.

10. Managing Storage Services + File Services

We will implement a robust storage solution that provides high availability, scalability, and data protection. This will include **RAID** (Redundant Array of Independent Disks) configurations for data redundancy and performance. We will also implement regular **backups** and disaster recovery procedures to ensure business continuity in the event of a failure. **File services** will be configured to provide secure access to files and folders based on user roles and permissions.

Why: Efficient storage management ensures data availability, supports scalability, and provides redundancy to protect against data loss.

Conclusion

Our plan for PromisedLand.com's new data center is all about creating a secure, flexible, and smooth-running system. We're using a mix of cloud and on-site technology, along with virtualization, to build a setup that can grow and adapt as needed. We're focusing on keeping things secure, making sure the network is dependable, storing data efficiently, and setting up strong security measures. This way, the new data center can handle the demands of today's call centers, while also being ready for whatever the future holds.

Part B

1.File + Storage Services

Step 1: Install the File and Storage Services Role

1. Open **Server Manager**.
2. Click on **Manage** and then select **Add Roles and Features**.
3. Follow the wizard until you reach the **Roles** section.
4. Check the box for **File and Storage Services**.
5. Continue through the wizard and complete the installation.

Step 2: Configure Storage Pools and Volumes

1. After installation, go back to **Server Manager**.
2. Click on **File and Storage Services** in the left pane.
3. Select **Storage Pools** and then click **Create Storage Pool**.
4. Follow the wizard to create a storage pool using the available disks.
5. Once the storage pool is created, click on **Create Virtual Disk** to create volumes.

Step 3: Set Up File Shares

1. In **Server Manager**, go to **File and Storage Services** > **Shares**.
2. Click **Create** and then **SMB Share**.
3. Follow the wizard to create a new SMB share, specifying the path, share name, and permissions.
4. Configure additional settings such as quotas, encryption, and access-based enumeration if needed.

Step 4: Configure NTFS Permissions

1. Right-click the file share you created and select **Properties**.
2. Go to the **Security** tab and click **Edit** to set NTFS permissions for different users or groups.
3. Assign appropriate permissions based on your organization's security policies.

2. Group Policy (Password & Audit Policies)

Step 1: Install the Group Policy Management Feature

1. **Open Server Manager**:
   - Click the **Start** button, then select **Server Manager**.
2. **Add Roles and Features**:
   - Click on **Manage** in the top-right corner, then select **Add Roles and Features**.
3. **Before You Begin**:
   - Click **Next** to proceed past the initial informational screen.
4. **Select Installation Type**:
   - Choose **Role-based or feature-based installation** and click **Next**.
5. **Select Destination Server**:
   - Select the server you want to install the feature on and click **Next**.
6. **Select Features**:
   - Scroll down and check the box for **Group Policy Management**.
   - Click **Next**, then **Install** to begin the installation.

Step 2: Open Group Policy Management Console (GPMC)

1. **Open GPMC**:
   - Press **Windows + R**, type `gpmc.msc`, and press **Enter** to open the Group Policy Management Console.

Step 3: Create a New GPO for Password and Audit Policies

1. **Create New GPO**:
   - In GPMC, expand your domain.
   - Right-click **Group Policy Objects** and select **New**.
   - Name the new GPO (e.g., "Password and Audit Policies") and click **OK**.
2. **Link GPO**:

- Right-click your domain (or a specific Organizational Unit (OU) if you want to apply it to specific users or computers) and select **Link an Existing GPO**.
- Choose the GPO you just created and click **OK**.

Step 4: Configure Password Policy

1. **Edit GPO**:
   - Right-click the new GPO and select **Edit**.
2. **Navigate to Password Policy**:
   - Go to **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **Account Policies** > **Password Policy**.
3. **Set Password Policies**:
   - **Enforce password history**: Set this to 24 passwords remembered.
   - **Maximum password age**: Set this to 60 days.
   - **Minimum password age**: Set this to 1 day.
   - **Minimum password length**: Set this to 12 characters.
   - **Password must meet complexity requirements**: Enable this policy.
   - **Store passwords using reversible encryption**: Disable this policy.

Step 5: Configure Account Lockout Policy

1. **Navigate to Account Lockout Policy**:
   - In the same GPO editor, go to **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **Account Policies** > **Account Lockout Policy**.
2. **Set Account Lockout Policies**:
   - **Account lockout duration**: Set this to 15 minutes.
   - **Account lockout threshold**: Set this to 5 invalid login attempts.
   - **Reset account lockout counter after**: Set this to 15 minutes.

Step 6: Configure Audit Policy

1. **Navigate to Audit Policy**:
   - Go to **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **Advanced Audit Policy Configuration** > **Audit Policies**.
2. **Set Audit Policies**:
   - Configure policies such as auditing account logon events, logon events, object access, etc., based on your organization's requirements.

4. DNS (Addresses, Zones, Forwarders, and Protection)

Step 1: Install the DNS Server Role

1. **Open Server Manager**:
   - Click the **Start** button, then select **Server Manager**.
2. **Add Roles and Features**:
   - Click on **Manage** in the top-right corner, then select **Add Roles and Features**.
3. **Before You Begin**:
   - Click **Next** to proceed past the initial informational screen.
4. **Select Installation Type**:
   - Choose **Role-based or feature-based installation** and click **Next**.
5. **Select Destination Server**:
   - Select the server you want to install the role on and click **Next**.
6. **Select Server Roles**:
   - Check the box for **DNS Server**.
   - Click **Add Features** when prompted, then click **Next**.
7. **Features**:
   - If there are any additional features you want to install, select them here. Otherwise, click **Next**.
8. **Confirmation**:
   - Review your selections and click **Install**. The installation process will begin, and you'll see a progress bar.

Step 2: Configure DNS Addresses

1. **Open DNS Manager**:
   - Press **Windows + R**, type `dnsmgmt.msc`, and press **Enter** to open the DNS Manager.
2. **Set Up Forward Lookup Zone**:
   - In the DNS Manager, right-click **Forward Lookup Zones** and select **New Zone**.
   - Follow the wizard to create a new primary zone.
   - Specify the zone name (e.g., `example.com`).
   - Choose to create a new zone file or use an existing one.
   - Complete the wizard to create the zone.
3. **Set Up Reverse Lookup Zone**:

- In the DNS Manager, right-click **Reverse Lookup Zones** and select **New Zone**.
- Follow the wizard to create a new primary zone.
- Specify the network ID for the reverse lookup zone.
- Complete the wizard to create the zone.

Step 3: Configure DNS Forwarders

1. **Open DNS Server Properties**:
   - In DNS Manager, right-click your DNS server and select **Properties**.
2. **Set Forwarders**:
   - Go to the **Forwarders** tab.
   - Click **Edit** and add the IP addresses of the DNS servers you want to use as forwarders (e.g., Google's DNS servers at `8.8.8.8` and `8.8.4.4`).
   - Click **OK** to save the settings.

Step 4: Configure DNS Zone Transfers and Replication

1. **Configure Zone Transfers**:
   - In DNS Manager, right-click the zone you created (e.g., `example.com`) and select **Properties**.
   - Go to the **Zone Transfers** tab.
   - Check **Allow zone transfers** and specify the IP addresses of the DNS servers allowed to receive the zone data.
   - Click **OK** to save the settings.
2. **Configure Zone Replication**:
   - In the same zone properties window, go to the **General** tab.
   - Set the **Replication** scope to specify which DNS servers will receive zone updates.

Step 5: Configure DNS Security (DNSSEC)

1. **Sign the Zone with DNSSEC**:
   - In DNS Manager, right-click the zone you want to secure and select **DNSSEC** > **Sign the Zone**.
   - Follow the wizard to configure DNSSEC settings and sign the zone.
2. **Configure Trust Anchors**:
   - Ensure that trust anchors are configured to validate DNSSEC signatures.

6. Network Policy Server

Step 1: Install the NPS Role

1. Open **Server Manager**.
2. Click on **Manage** (top-right corner) → Select **Add Roles and Features**.
3. In the **Before You Begin** page, click **Next**.
4. **Select Installation Type**: Choose **Role-based or feature-based installation**, then click **Next**.
5. **Select Destination Server**: Choose your server and click **Next**.
6. **Select Server Roles**:
   - Scroll down and check **Network Policy and Access Services**.
   - Click the **Add Features** button when prompted.
   - Click **Next**.
7. **Select Features**: No extra features are needed. Click **Next**.
8. **Confirm Installation**: Click **Install** and wait for the installation to complete.
9. Click **Close** when finished.

Step 2: Register NPS in Active Directory

After installing NPS, you must register it in **Active Directory** if it will authenticate domain users.

1. Open **Server Manager**.
2. Click **Tools** (top-right corner) → Select **Network Policy Server**.
3. In the NPS console, **right-click NPS (Local)** → Select **Register Server in Active Directory**.
4. Click **OK** in the confirmation dialog.

Step 3: Add a RADIUS Client (VPN Server, AP, or Router)

1. In the **NPS Console**, expand **RADIUS Clients and Servers**.
2. Right-click **RADIUS Clients** → Click **New**.
3. Fill in the details:
   - **Friendly Name**: Example: `VPN Server`
   - **Address (IP or DNS)**: Enter the IP address of the device (e.g., VPN server or AP).
   - **Shared Secret**: Click **Generate** or enter a secure key.

4. Click **OK** to save.

Step 4: Create a Network Policy

1. In the **NPS Console**, expand **Policies** → Right-click **Network Policies** → Click **New**.
2. **Policy Name**: Example: `Allow VPN Users`.
3. **Policy Type**: Select **Network Access Permission**.
4. Click **Next**.
5. **Specify Conditions**:
   - Click **Add** → Select **Windows Groups** → Click **Add Groups**.
   - Enter `VPNUsers` (or another AD group).
   - Click **OK**.
6. Click **Next**.
7. **Grant Access**: Select **Grant Access**, then **Next**.
8. **Authentication Methods**:
   - Ensure **MS-CHAP v2** is checked (for VPN).
   - Click **Next**.
9. Click **Next** through the configuration screens and then **Finish**.

Step 5: Configure Connection Request Policy

1. Expand **Policies** → Right-click **Connection Request Policies** → Click **New**.
2. **Policy Name**: Example: `VPN Connection Request`.
3. **Specify Conditions**:
   - Click **Add** → Select **NAS Port Type** → Choose **Virtual (VPN)**.
   - Click **OK**.
4. Click **Next** → Select **Authenticate requests on this server** → Click **Next**.
5. Click **Finish**.

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7. Data Protection

Step 1: Configure Backup

1. **Install Windows Server Backup**:
    - Open **Server Manager**.
    - Click on **Manage** and then select **Add Roles and Features**.
    - Follow the wizard until you reach the **Features** section.
    - Check the box for **Windows Server Backup** and complete the installation.
2. **Set Up Backup Schedule**:
    - Open **Windows Server Backup** from the Tools menu in Server Manager.
    - Click **Local Backup**, then **Backup Schedule**.
    - Follow the wizard to create a backup schedule, specifying the items to back up, the backup destination, and the schedule frequency.
3. **Run a One-Time Backup** (Optional):
    - In Windows Server Backup, click **Local Backup**, then **Backup Once**.
    - Follow the wizard to configure a one-time backup.

Step 2: Implement BitLocker for Disk Encryption

1. **Open BitLocker Management**:
    - Press **Windows + R**, type `gpedit.msc`, and press **Enter** to open the Group Policy Editor.
2. **Configure BitLocker Settings**:
    - Navigate to **Computer Configuration** > **Administrative Templates** > **Windows Components** > **BitLocker Drive Encryption**.
    - Configure settings such as **Choose drive encryption method and cipher strength**, **Require additional authentication at startup**, etc.
3. **Enable BitLocker**:
    - Open **Control Panel** > **System and Security** > **BitLocker Drive Encryption**.
    - Click **Turn on BitLocker** for the desired drive and follow the wizard to complete the encryption process.

Step 3: Set Up File and Folder Permissions

1. **Configure NTFS Permissions**:
    - Right-click the file or folder you want to protect and select **Properties**.
    - Go to the **Security** tab and click **Edit**.
    - Add or modify the permissions for users or groups, specifying the allowed or denied actions.
2. **Implement Access Control Lists (ACLs)**:

- Open **Command Prompt** with administrative privileges.

Use the `icacls` command to configure ACLs for files and folders. For example:
plaintext
icacls "C:\YourFolder" /grant YourUser:(OI)(CI)(F)

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Step 4: Enable File and Folder Auditing

1. **Open Group Policy Management Console (GPMC)**:
   - Press **Windows + R**, type `gpmc.msc`, and press **Enter**.
2. **Create or Edit a GPO**:
   - Create a new Group Policy Object (GPO) or edit an existing one.
3. **Configure Audit Policy**:
   - Navigate to **Computer Configuration** > **Policies** > **Windows Settings** > **Security Settings** > **Advanced Audit Policy Configuration** > **Audit Policies** > **Object Access**.
   - Enable auditing for **Audit File System** and specify the desired audit settings (e.g., Success, Failure).
4. **Set Auditing for Specific Files or Folders**:
   - Right-click the file or folder you want to audit, select **Properties**, and go to the **Security** tab.
   - Click **Advanced**, then go to the **Auditing** tab.
   - Add a new auditing entry and specify the users or groups to audit, along with the actions to audit (e.g., Read, Write).

Step 5: Monitor and Manage Data Protection

1. **Review Backup Logs**:
   - Open **Event Viewer** and navigate to **Applications and Services Logs** > **Microsoft** > **Windows** > **Backup** to review backup logs and identify any issues.
2. **Monitor BitLocker Status**:
   - Use **BitLocker Drive Encryption** in Control Panel to check the status of encrypted drives.
3. **Analyze Audit Logs**:
   - Open **Event Viewer** and navigate to **Windows Logs** > **Security** to review audit logs and monitor access to protected files and folders.

8. Performance Monitoring

Step 1: Open Performance Monitor

1. Open **Server Manager**.
2. Click **Tools** (top-right corner) → Select **Performance Monitor**.
3. The **Performance Monitor Console** opens, showing a basic real-time graph.

Step 2: Add Performance Counters

To monitor system performance in real-time:

1. In the **Performance Monitor** window, expand **Monitoring Tools**.
2. Click **Performance Monitor**.
3. Click the **green plus (+) button** on the toolbar.
4. In the **Add Counters** window:
    - Select a counter category (e.g., **Processor**, **Memory**, **Disk**, **Network**).
    - Choose a specific counter (e.g., **% Processor Time** under **Processor**).
    - Click **Add >>**.
5. Click **OK** to start monitoring.

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